Crypto Lab 1 - Shishir Vuppala

Challenge 4: The Catastrophic Equality

flag: `cs409{y0u_ch3ck3d_th3_wr0ng_s1d3_0f_th3_c1ph3r}`

Solution

Because the same IV is reused every time we try to decrypt, we can use a specially crafted ciphertext to recover it. Let the block size be N, let C be any ciphertext block and let 0^N denote a block of only zeros. So, we send the ciphertext $C\|0^N\|C$ to the server. let's look at how the decryption equations look like.

$$P_i = \text{Dec}(C_i) \oplus C_{i-1} \text{ for i from 1 to blockcount and } C_0 = \text{IV}.$$

For our ciphertext we have $C_0 = \text{IV}; C_1 = C; C_2 = 0^N$ and $C_3 = C$

$$P_1 = \text{Dec}(C) \oplus \text{IV}$$

$$P_2 = \text{Dec}(0^N) \oplus C$$

$$P_3 = \text{Dec}(C) \oplus 0^N = Dec(C)$$

Thus we have

$$P_1 \oplus P_3 = (\text{Dec}(C) \oplus \text{IV}) \oplus \text{Dec}(C) = \text{IV}$$

Now, because the IV is also the key, we can use the IV to recreate the same cipher in our code and then use it to encrypt 'admin=true' and then send the generated ciphertext to the server to get the encrypted flag. We can then decrypt it by using the same recovered key/IV.

Code

The script used to solve this challenge is written in `solution_template.py`